



Linux & WSL Command Reference

Follow in order: install → navigate → files → system → network → git → fun

NEW helpful addition **MUST** must-know **SAFE** no harm **WOW** cool / impressive

1 Install WSL

WHERE	COMMAND	WHAT IT DOES
Windows	<code>wsl --install</code>	Install WSL + default Ubuntu MUST
Windows	<code>wsl --install -d Ubuntu-24.04 --location D:\WSL\Ubuntu</code>	Install Ubuntu directly to D: drive NEW
Windows	<code>wsl --install -d Ubuntu-24.04</code>	Install a specific Ubuntu version
Windows	<code>wsl -l -v</code>	List installed distros + WSL version MUST
Windows	<code>wsl --status</code>	Show WSL status
Windows	<code>wsl --update</code>	Update WSL
Windows	<code>wsl --shutdown</code>	Stop all WSL MUST

After install: reboot → set username + password. Your Linux password is only needed with `sudo`.

2 Navigate the Filesystem

COMMAND	WHAT IT DOES
<code>pwd</code>	Where am I? MUST
<code>ls</code>	List files MUST
<code>ls -la</code>	List all (incl. hidden) with details MUST
<code>ls -lh</code>	List with human-readable sizes (KB/MB) NEW
<code>ls -lt</code>	List sorted newest first NEW
<code>cd <folder></code>	Go into folder MUST
<code>cd ..</code>	Go up one level MUST
<code>cd ~</code>	Go home MUST
<code>cd /</code>	Go to root of Linux
<code>cd /mnt/c</code>	Go to Windows C: drive
<code>cd /mnt/d</code>	Go to Windows D: drive
<code>tree -L 2</code>	Visual folder tree (2 levels deep) WOW

WSL is case-sensitive: Newfolder ≠ newfolder. **Install tree:** `sudo apt install tree`

3 Create & Edit Files

COMMAND	WHAT IT DOES
<code>touch file.txt</code>	Create empty file
<code>mkdir folder</code>	Create a folder
<code>mkdir -p a/b/c</code>	Create nested folders at once NEW
<code>nano file.txt</code>	Simple text editor (Ctrl+X to save/exit) MUST
<code>code file.txt</code>	Open file in VS Code
<code>cat file.txt</code>	View entire file
<code>head -20 file.txt</code>	Show first 20 lines
<code>tail -20 file.txt</code>	Show last 20 lines
<code>tail -f log.txt</code>	Live-follow a log (Ctrl+C to stop) WOW

4 Copy, Move, Rename, Delete

COMMAND	WHAT IT DOES
<code>cp file.txt /path/</code>	Copy a file
<code>cp -r folder/ /path/</code>	Copy folder + everything inside MUST
<code>mv old.txt new.txt</code>	Rename file
<code>mv file.txt /path/</code>	Move file
<code>rm file.txt</code>	Delete file
<code>rm -ri folder</code>	Delete folder with confirmation (safe!) SAFE
<code>cp -r /mnt/c/MyFolder /mnt/d/</code>	Copy folder from C: to D: NEW

`wc -l file.txt` Count lines in file **NEW**

Tip: `-r` = recursive (required for folders). Always `rm -ri` first time — it's safe!

5 Search & View Files

COMMAND	WHAT IT DOES
<code>grep "word" file.txt</code>	Search inside a file
<code>grep -r "text" .</code>	Search recursively in all files WOW
<code>find . -name "*.txt"</code>	Find all .txt files
<code>find . -size +10M</code>	Find files bigger than 10 MB NEW
<code>diff a.txt b.txt</code>	Compare two files NEW
<code>sort names.txt</code>	Sort lines alphabetically NEW
<code>uniq sorted.txt</code>	Remove duplicate lines NEW

6 System Info

COMMAND	WHAT IT DOES
<code>uname -r</code>	Kernel version
<code>lsb_release -a</code>	Ubuntu version NEW
<code>cat /etc/os-release</code>	Full OS info
<code>whoami</code>	Current user MUST
<code>id</code>	User ID + groups NEW
<code>free -h</code>	RAM usage
<code>df -h</code>	Disk space
<code>df -h /mnt/c</code>	Real SSD free space MUST
<code>du -sh *</code>	Size of each item here
<code>du -sh /path</code>	Size of specific folder NEW
<code>top</code>	CPU + memory live (q to quit) MUST
<code>nproc</code>	Number of CPU cores
<code>uptime</code>	How long system running NEW
<code>date</code>	Show current date & time

WSL "1024 GB" is virtual disk size, not your SSD. Check real space with `df -h /mnt/c`.

7 Install Software (Packages)

COMMAND	WHAT IT DOES
<code>sudo apt update</code>	Refresh package list MUST
<code>sudo apt upgrade</code>	Upgrade all packages MUST
<code>sudo apt install <name></code>	Install software (e.g. git, http)
<code>sudo apt remove <name></code>	Uninstall software
<code>sudo apt autoremove</code>	Clean up unused packages NEW
<code>sudo apt search <keyword></code>	Search for a package NEW
<code>which <command></code>	Find where a command is installed NEW

8 Permissions

COMMAND	WHAT IT DOES
<code>sudo <command></code>	Run as admin (asks password) MUST
<code>chmod +x script.sh</code>	Make file executable
<code>chmod 755 file</code>	Full access owner, read+execute others NEW
<code>chown user file</code>	Change file owner NEW

Password only asked with **sudo** — not when you just open the terminal.

9 Process Management

COMMAND	WHAT IT DOES
<code>top</code>	Live process monitor (q to quit)
<code>htop</code>	Fancier top with colours WOW
<code>ps aux</code>	List all running processes
<code>ps aux grep node</code>	Find a specific process by name NEW
<code>kill <PID></code>	Stop a process
<code>kill -9 <PID></code>	Force-kill a stuck process NEW
<code>nohup cmd &</code>	Run in background (survives close) WOW

Install htop: `sudo apt install htop` — much better than top!

10 Networking

COMMAND	WHAT IT DOES
<code>ping google.com</code>	Test internet connection
<code>curl -I https://example.com</code>	Check HTTP headers NEW
<code>wget https://url/file.zip</code>	Download a file NEW
<code>ip addr</code>	Show your IP addresses NEW
<code>ss -tlnp</code>	Show which ports are in use WOW
<code>nslookup google.com</code>	DNS lookup NEW
<code>traceroute google.com</code>	Trace path to a website WOW

11 Git (Version Control)

12 GUI & Open in Windows

COMMAND	WHAT IT DOES
git init	Start a new repo
git status	See what's changed MUST
git add .	Stage all changes
git commit -m "msg"	Save a snapshot
git log --oneline	Commit history (short) NEW
git branch	List branches NEW
git branch <name>	Create a branch NEW
git checkout <name>	Switch branch NEW
git clone <url>	Download a repo NEW

Install: sudo apt install git

COMMAND	WHAT IT DOES
explorer.exe .	Open folder in Windows Explorer
notepad.exe file.txt	Open file in Windows Notepad NEW
code .	Open folder in VS Code
sudo apt install nautilus	Install Linux file manager
nautilus .	Open Linux file manager

Nautilus display error? Run **wsl --update** then **wsl --shutdown** from Windows.

13 Shell Shortcuts & History

COMMAND	WHAT IT DOES	SHORTCUT	ACTION
history	Show all typed commands WOW	Tab	Autocomplete
history grep python	Search your history WOW	Ctrl+C	Stop running command
!!	Re-run the last command WOW	Ctrl+R	Search history
clear / Ctrl+L	Clear screen MUST	↑ ↓	Previous / next commands
alias ll="ls -la"	Make a shortcut command NEW	Ctrl+A / Ctrl+E	Jump to start / end of line
exit	Close WSL terminal	Ctrl+Shift+V	Paste

14 Combining Commands (Pipes)

PATTERN	WHAT IT DOES
cmd1 cmd2	Pipe: output of cmd1 → cmd2 MUST
cmd1 && cmd2	Run cmd2 only if cmd1 works NEW
ls wc -l	Count files in folder WOW
ps aux grep node	Find all node processes
history tail -20	Last 20 commands you typed
ls -lh grep ".pdf"	Show only PDF files NEW
du -sh * sort -hr	Folders biggest → smallest WOW
cat f sort uniq	Sort + remove duplicates

Pipes | are Linux's superpower — chain as many as you want!

15 Quick Scripting

COMMAND	WHAT IT DOES
echo "hello"	Print text to screen

16 Cool Fun Commands

COMMAND	WHAT IT DOES
cal	Show calendar WOW
fortune	Random fun quote WOW

echo "hi" > file.txt	Write to file (overwrites!) NEW
echo "hi" >> file.txt	Append to file NEW
env grep PATH	Show environment variables NEW
sleep 5	Wait 5 seconds
seq 1 10	Print numbers 1 to 10 WOW
for i in \$(seq 1 5); do echo \$i; done	Quick loop! WOW

Quick script file:

```
#!/bin/bash
echo "Hello $USER! Today is $(date)"
if [ -d "node_modules" ]; then
  echo "node_modules exists"
fi
```

Save: nano mystart.sh → chmod +x mystart.sh → run: ./mystart.sh

fortune cowsay	Cow speaks a random quote WOW
cowsay "hello"	Cow says your text! WOW
cowsay -f dragon "hi"	Dragon says hi WOW
cowsay -f tux "Linux!"	Tux (penguin) says text WOW
cmatrix	Matrix falling letters (Ctrl+C to stop) WOW
sl	Train animation (if you miss "ls!") WOW
figlet "HELLO"	Big ASCII text banner WOW
toilet "HELLO"	Fancy coloured ASCII banner WOW
yes	Prints "y" forever (Ctrl+C to stop) WOW
yes "lol"	Prints "lol" forever WOW
watch -n 2 free -h	Auto-update command every 2 sec WOW
factor 42	Prime factorisation WOW
bc -l	Interactive calculator (pi, sin, etc.) WOW
echo "scale=5; 22/7" bc	Quick calculator — get pi! WOW
expr 2 + 2	Quick math in terminal NEW
shuf -i 1-100 -n 1	Random number between 1-100 WOW
echo \$RANDOM	Random number (0-32767) NEW
strings /dev/urandom head -1	Random gibberish text WOW
date +%A	What day is today? NEW
uptime -p	Human-readable uptime NEW
date +%s	Unix timestamp NEW
time <cmd>	Measure command speed NEW

Install fun pack: sudo apt install fortune cowsay cmatrix sl figlet toilet bc

17 WSL Management (Windows PowerShell)

WSL Basics

COMMAND	WHAT IT DOES
wsl --install	Install WSL + Ubuntu
wsl -l -v	List distros + WSL version
wsl --status	Show WSL status
wsl --update	Update WSL
wsl --shutdown	Stop all WSL
wsl --set-version Ubuntu 2	Switch to WSL2 NEW
wsl -u root	Root login (reset password) NEW

Move / Backup / Uninstall

COMMAND	WHAT IT DOES
<code>wsl --install -d Ubuntu --location D:\WSL\Ubuntu</code>	Install to D: drive NEW
<code>wsl --export Ubuntu D:\backup.tar</code>	Back up a distro NEW
<code>wsl --unregister Ubuntu</code>	Remove a distro
<code>wsl --uninstall</code>	Uninstall WSL
<code>dism ... /disable-feature ...Subsystem-Linux</code>	Disable WSL feature (Admin)
<code>dism ... /disable-feature ...VirtualMachinePlatform</code>	Disable VM platform (Admin)

After **wsl --uninstall**: reboot, then delete leftover folders on D:\WSL and in %LOCALAPPDATA%\Packages*Ubuntu*

18 Comments in Scripts

SYNTAX	WHAT IT DOES
<code># this is a comment</code>	Ignored by the shell — used for notes in scripts
<code>echo "Hi" # end-of-line comment</code>	Comment after a command on the same line

Colour scheme designed for colour-blind accessibility · Use **Cmd/Ctrl+F** to search any command · Steps 1 → 18 in learning order